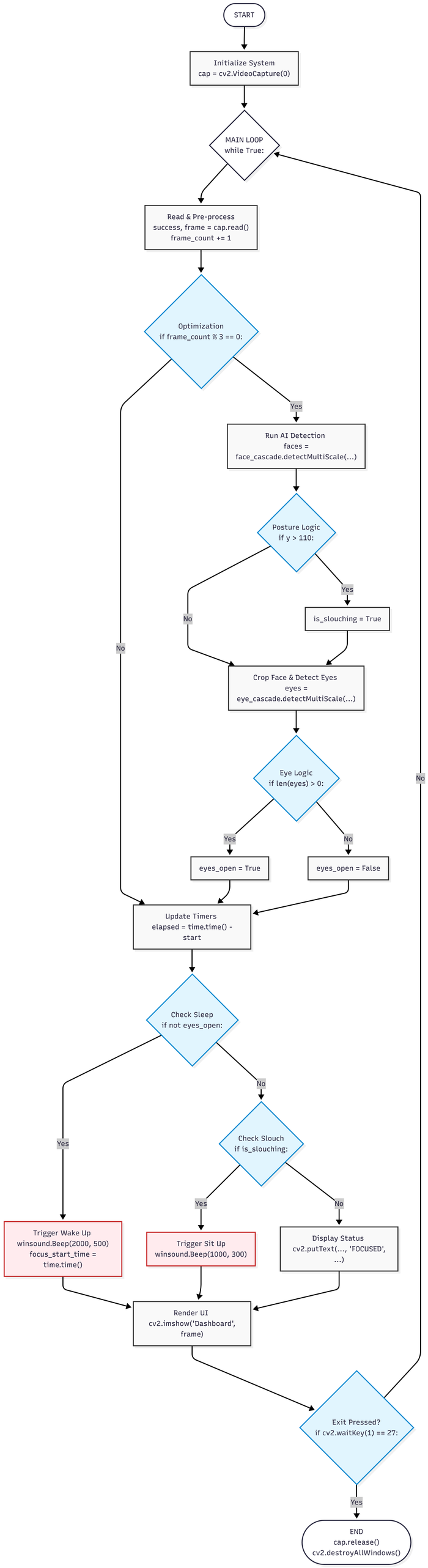
START
Initialize System
cap = cv2.VideoCapture(0)
MAIN LOOP
while True:
Read & Pre-process
success, frame = cap.read()
frame_count += 1
Optimization
if frame_count % 3 == 0:
Yes
No
Run AI Detection
faces =
face_cascade.detectMultiScale(...)
Posture Logic
if y > 110:
Yes
No
is_slouching = True
Crop Face & Detect Eyes
eyes =
eye_cascade.detectMultiScale(...)
Eye Logic
if len(eyes) > 0:
Yes
No
eyes_open = True
eyes_open = False
Update Timers
elapsed = time.time() -
start
Check Sleep
if not eyes_open:
Yes
No
Check Slouch
if is_slouching:
Yes
No
Trigger Wake Up
winsound.Beep(2000, 500)
focus_start_time =
time.time()
Trigger Sit Up
winsound.Beep(1000, 300)
Display Status
cv2.putText(..., 'FOCUSED',
...)
Render UI
cv2.imshow('Dashboard',
frame)
Exit Pressed?
if cv2.waitKey(1) == 27:
No
Yes
END
cap.release()
cv2.destroyAllWindows()